How-To Library

Informatica

How to Set Up Informatica Managed File Transfer Gateway for Forward and Reverse Proxy

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Abstract

Informatica Managed File Transfer Gateway allows trading partners a secure connection to your organization without keeping any sensitive information in the demilitarized zone (DMZ) or opening incoming ports into your private network. Managed File Transfer Gateway integrates with the Managed File Transfer file server to provide a secure and managed solution for hosting and exchanging files. This article provides detailed information about how to set up Informatica Managed File Transfer Gateway for forward and reverse proxy.

Supported Versions

- B2B Data Exchange 10.2.2

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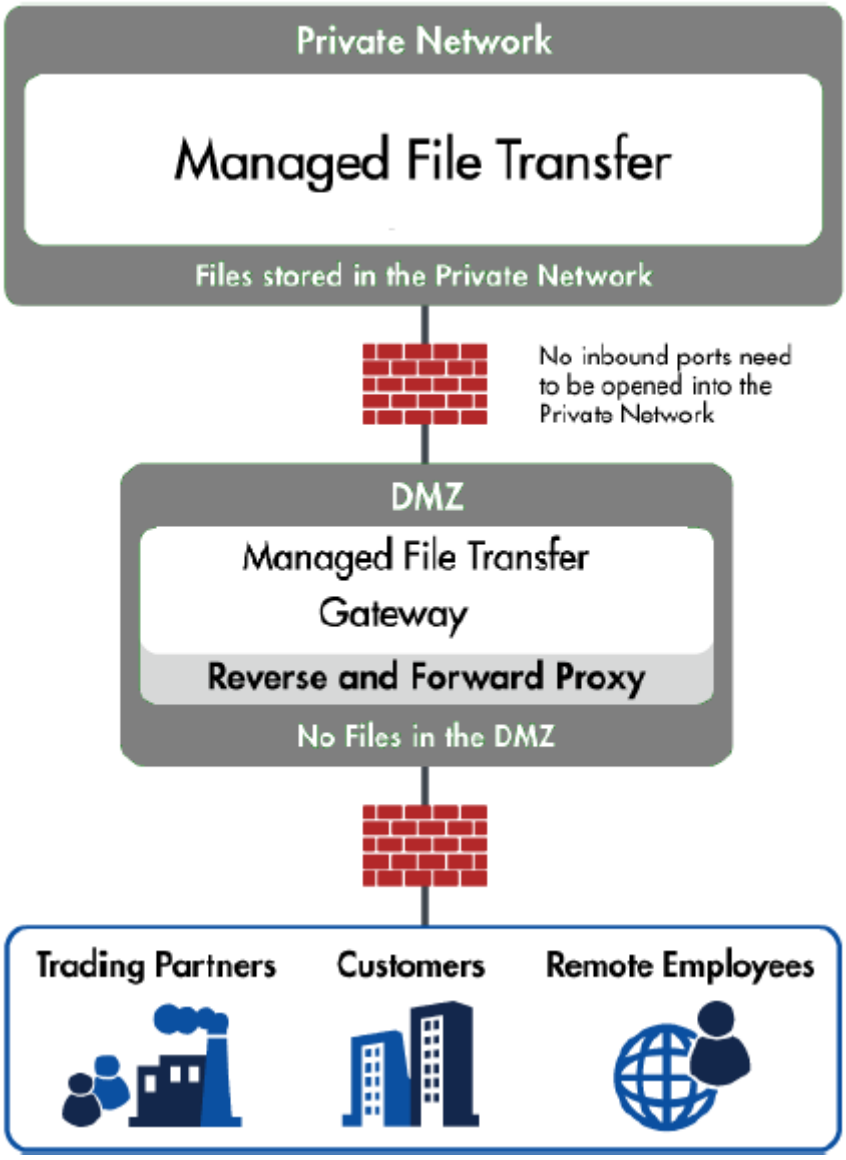
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Overview

Informatica Managed File Transfer Gateway allows trading partners a secure connection to your organization without storing any sensitive information in the DMZ or opening incoming ports into your private network. Managed File

Transfer Gateway integrates with the Managed File Transfer file server to provide a secure and managed solution for hosting and exchanging files.

The following image shows a model of secure communications using the Managed File Transfer Gateway:



Introduction

Informatica Managed File Transfer Gateway includes the following features:

- Load balancing when multiple Managed File Transfer systems are running in a clustered configuration.
- Incoming ports are not opened into the private network.
- Files are not stored in the DMZ.
- User credentials and permissions are maintained and stored on the private network.
- Service configurations are maintained and stored on the private network .
- Supports FTP/s, SFTP, HTTPs, and MLLP file transfer protocols.

- Available as a software-only solution.
- Installs on Windows, Linux, AIX, UNIX, and Solaris operating systems.

Setting Up Managed File Transfer Gateway for Reverse Proxy

You can configure Informatica Managed File Transfer Gateway as a reverse proxy. When you use Informatica Managed File Transfer Gateway as a reverse proxy, the Informatica Managed File Transfer Gateway does not open the inbound ports into the internal network and does not store sensitive data in the DMZ.

Perform the following steps to set up Informatica Managed File Transfer Gateway as a reverse proxy:

1. Add a new Gateway configuration.
2. Configure the Gateway parameters.

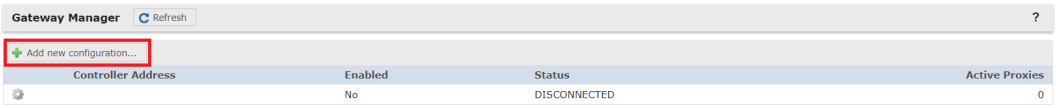
You can also perform a Gateway Manager action to view, edit, copy, or delete a Gateway.

Gateway Manager

Use the Gateway Manager to control the Managed File Transfer Gateway connection and view the current status of the Managed File Transfer Gateway. When Managed File Transfer is running in a cluster configuration, the service mappings for each system in the cluster are displayed in the Gateway Manager.

Perform the following steps to add a new Gateway configuration:

1. To manage the Managed File Transfer Gateway, log in as an Admin User with the **Product Administrator** role.
2. From the main menu bar, select **Services** and then click **Gateway Manager**.
   The following image shows the **Gateway Manager** page:



3. Add a new Gateway configuration by clicking the **Add new configuration** link.

Gateway Configuration

The Gateway Configuration page contains the operating parameters and service mappings for Managed File Transfer Gateway.

Configure the Gateway parameters and service mappings for Managed File Transfer Gateway.

The following table describes the Gateway Configuration parameters:

Gateway Configuration parameter	Description
Enabled	Indicates whether Informatica Managed File Transfer Gateway is enabled. By default, Informatica Managed File Transfer Gateway is disabled.
Automatically Start	Indicates whether Informatica Managed File Transfer Gateway connection is started automatically when Informatica Managed File Transfer starts.

Gateway Configuration parameter	Description
Controller Address	The IP address of the server running the Informatica Managed File Transfer Gateway server component. The IP address must match the controller address defined in the `gateway.xml` file on the Informatica Managed File Transfer Gateway server.
Controller Port	The port on which Informatica Managed File Transfer Gateway listens for incoming control connections. The default port is 9100 and must match the port defined in the `gateway.xml` file on the Informatica Managed File Transfer Gateway Server.
Minimum Number of Threads	The minimum number of threads reserved for connections to Informatica Managed File Transfer Gateway. Default is 10.
Maximum Number of Threads	The maximum number of threads that can be used for connections to Informatica Managed File Transfer Gateway. This determines the maximum number of simultaneous requests that can be handled. Default is 500. **Note:** Changing the thread values can alter the performance. Modify the thread values to obtain the optimal performance for your configuration requirements.
Thread Keep Alive Time	The amount of time that an idle thread waits before it is released. Default is 60 seconds. **Note:** Increasing the value might cause a higher CPU and memory usage on the server. However, the connections start faster because they do not need to create new threads frequently.

Gateway Configuration parameter	Description
Service Mappings	Each service such as IFTP, FTPS, HTTPS, and SFTP is mapped from Informatica Managed File Transfer Gateway to the corresponding service in Informatica Managed File Transfer. When a control connection is established, the mappings are sent to the Informatica Managed File Transfer Gateway server. Informatica Managed File Transfer Gateway listens for incoming connections on the **From Address** and **From Port** for each service mapping. When a connection occurs, Informatica Managed File Transfer Gateway forwards the traffic to Informatica Managed File Transfer based on the **To Address** and **To Port** service mappings. If more service mappings are needed, for example, more than one listener for a particular service, click the **Add Service Mapping** link to add a new service mapping entry row. If a service mapping is no longer needed, click the icon to delete the corresponding service mapping.
Add service mapping	The Add service mapping contains the following parameters: **Label** An identifier for the service mapping. **From Address** The IP address that Managed File Transfer Gateway uses to listen on for incoming traffic. Optionally leave the From Address field blank to use the Controller Address as the bind address for client connections on that particular mapping. **From Port** The port that Managed File Transfer Gateway uses to listen on for incoming connections. The standard ports are 21 for FTP, 990 for FTPS, 22 for SFTP, and 443 for HTTPS/AS2. **To Address** The IP address of the local Managed File Transfer system. **To Port** The port that the service listener uses for a specific service on the Managed File Transfer system. **Load Balancer Rule** The Managed File Transfer Gateway uses the Load Balancer Rule to configure the load balancing for each service mapping. This rule is defined in Managed File Transfer Gateway and should be round-robin for FTP, FTPS, and SFTP protocols. For HTTPS, the Load Balancer Rule should be IP-based round-robin that routes traffic from specific IP addresses for a period of time to a single system. By default, use the `default` rule for FTP, FTPS and SFTP. Use the `https` rule for HTTPs and AS2. Changes made to the Service Mappings take effect the next time the connection to the Managed File Transfer Gateway is reset. To reset the connection, click the **Disconnect** button and then the **Connect** button on the **Gateway Manager** page. For more information, see Gateway Manager.

The following image shows the **Gateway Configuration** page:

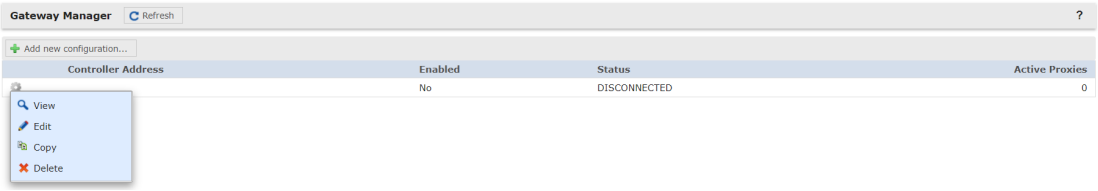


Perform a Gateway Manager Action

You can perform the following actions by selecting the Actions icon:

- View the Gateway details by clicking the icon.
- Edit a Gateway by clicking the icon.
- Copy a Gateway by clicking the icon.
- Delete a Gateway by clicking the icon.

The following image shows the Gateway Manager actions:



Setting Up Informatica Managed File Transfer Gateway for Forward Proxy

You can configure Informatica Managed File Transfer Gateway as a forward proxy. When you use Informatica Managed File Transfer Gateway as a forward proxy for outbound connections, the Informatica Managed File Transfer Gateway hides the identities and locations of the internal systems.

Perform the following actions to set up Informatica Managed File Transfer Gateway as a forward proxy:

- Configure the proxy tab for AS2 resource.
- Configure the proxy tab for MLLP resource.
- Configure the proxy tab for STFP resource.

Configure the Proxy Tab for AS2 Servers Resource

The AS2 Server Resource is used to specify the settings to use when messages are sent using the *AS2* 1.2 specification. AS2 is a standard originally created to securely transfer EDI documents, but it can also be used to

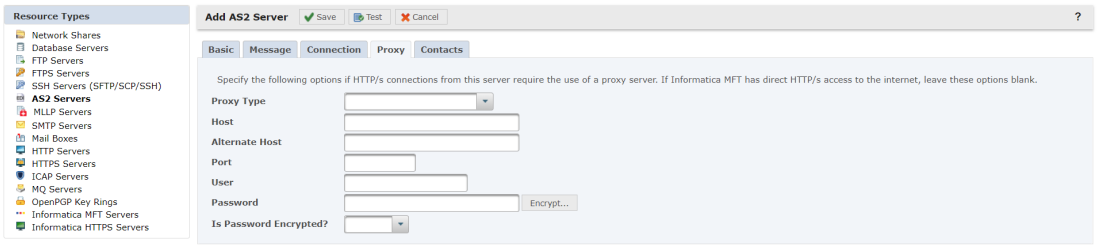
transmit virtually any file type. The messages are structured using the standard S/MIME format and are sent over HTTP(S) connections.

Configure the following fields in the **Proxy Tab** to set up Informatica Managed File Transfer Gateway for forward proxy:

Proxy Tab field	Description
Proxy Type	Managed File Transfer supports SOCKS (version 4 and 5), HTTP tunneling through an HTTP proxy, and Managed File Transfer Gateway. Check with the network administrator for the correct proxy type.
Host	The host name or IP address of the proxy server on your network.
Alternate Host	The host name or IP address of an alternate proxy server. The alternate proxy server is used when the primary proxy server is unavailable.
Port	The port number to use for connecting to the proxy server. The default port for an HTTP connection is 80 and SOCKS is 1080.
User	The user name to use to connect to the proxy server.
Password	The password to use to connect to the proxy server. Required if your network uses a proxy server to make HTTP(S) connections. After entering the password, you can optionally click the **Encrypt** button which will encrypt the password when it is stored in the Managed File Transfer database.
Is Password Encrypted	Indicates whether the password is encrypted. Choose **Yes** if you click the **Encrypt** button for the Password.

Note: The options are only needed if your system uses a proxy server to make HTTP(S) connections.

The following image shows the fields that you need to configure in the **Proxy** tab:



Configure the Proxy Tab for MLLP Servers Resource

The Minimal Lower Layer Protocol (MLLP) protocol is used to transfer healthcare industry messages, such as HL7 messages. HL7 is a messaging specification for healthcare information systems.

The MLLP protocol is a minimalistic OSI-session layer framing protocol. HL7 is a messaging specification for healthcare information systems. The Managed File Transfer resource can connect to an MLLP server to send files. When defining an MLLP resource in Managed File Transfer, you need to indicate the connection properties, such as the host name or IP address, and port number.

Configure the following fields in the **Proxy Tab** to set up Informatica Managed File Transfer Gateway for forward proxy:

Proxy Tab field	Description
Proxy Type	Managed File Transfer supports SOCKS (version 4 and 5), HTTP tunneling through an HTTP proxy, and Managed File Transfer Gateway. HTTP tunneling requires that the HTTP proxy supports the CONNECT HTTP method. Not all HTTP proxy servers may support the CONNECT method and some might only allow HTTPS traffic. When using an HTTP proxy that requires authentication, Basic and Digest authentication schemes are supported. Check with the network administrator for the correct proxy type.
Host	The host name or IP address of the proxy server. **Note:** If the Proxy Type or Host fields are blank, there is a direct connection to the target host.
Alternate Host	The host name or IP address of an alternate proxy server. The alternate proxy server is used when the primary proxy server is unavailable.
Port	The port number to use for connecting to the proxy server. The default port for an HTTP connection is 80 and SOCKS is 1080.
User	The user name to use for connecting to the proxy server.
Password	The password to use for connecting to the proxy server. After entering the password, you can optionally click the **Encrypt** button which will encrypt the password when it is stored in the Managed File Transfer database. **Note:** If you do not want to store the password for the proxy server in the FTP server resource, enter the password when you execute a Project.
Is Password Encrypted	Indicates whether the password is encrypted. Choose **Yes** if you click the **Encrypt** button for the Password.

Configuring the Proxy Tab for SFTP Servers Resource

Informatica Managed File Transfer can connect to SSH Servers for performing SFTP transfers, SCP (Secure Copy) file transfers, and for running SSH remote commands. When you define an SSH Server resource in Managed File Transfer, indicate the connection properties such as the host name or IP address, and User ID. You can specify a password, SSH private key, or both for authentication.

Configure the following fields in the **Proxy Tab** to set up Informatica Managed File Transfer Gateway for forward proxy:

Proxy Tab field	Description
Proxy Type	Managed File Transfer supports SOCKS (version 4 and 5), HTTP tunneling through an HTTP proxy, and Managed File Transfer Gateway. HTTP tunneling requires that the HTTP proxy supports the CONNECT HTTP method. Not all HTTP proxy servers may support the CONNECT method and some might only allow HTTPS traffic. When using an HTTP proxy that requires authentication, Basic and Digest authentication schemes are supported. Check with the network administrator for the correct proxy type.
Host	The host name or IP address of the proxy server. **Note:** If the Proxy Type or Host fields are blank, there is a direct connection to the target host.

Proxy Tab field	Description
Alternate Host	The host name or IP address of an alternate proxy server. The alternate proxy server is used when the primary proxy server is unavailable.
Port	The port number to use for connecting to the proxy server. The default port for an HTTP connection is 80 and SOCKS is 1080.
User	The user name to use for connecting to the proxy server.
Password	The password to use for connecting to the proxy server. After you enter the password, you can optionally click the **Encrypt** button that will encrypt the password when it is stored in the Managed File Transfer database.
Is Password Encrypted	Indicates whether the password is encrypted. Choose **Yes** if you click the **Encrypt** button for the Password.

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